

Exploring Narratives in Swiss Media

A Structured NLP Pipeline to Extract, Cluster and Analyze Macro-, Meso- and Micro-Narratives across Different Media Outlets

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Abstract

This project presents an end-to-end natural language processing (NLP) pipeline that extracts the narrative structure of news articles, clusters those structures across texts, and thereby surfaces the dominant narratives in Swiss political and economic reporting. Starting from 245,080 articles retrieved from the Swissdox database, the corpus is filtered to 66'404 politics- and economy-related articles published between 2025-06-01 and 2026-05-31 across 12 German-language Swiss outlets (7 publisher groups).

Each article is analyzed by a large language model into a Greimas-style actantial structure — subject, object, sender, receiver, helper and opponent — whose fillers are typed against a three-level societal-actor taxonomy.

A narrative was found in 67.4% of articles (44'786 articles). Embedding these structured narratives with a multilingual sentence encoder and clustering them with UMAP + HDBSCAN yields 17 macro-, 226 meso- and 351 micro-narratives. An interactive Streamlit application makes the full narrative landscape explorable.

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1. Introduction

In recent years, many news media have increasingly come under scrutiny for the way in which they propagate, frame or omit particular political narratives within their reporting. Since the public's perception of, trust in and normative evaluation of political actors, events and processes are to no small extent shaped through how such elements are portrayed in public discourse, examining the prevalence of different narrative frames in the news is a relevant task for contemporary political analysis.

With the advent of large language models (LLMs) and, in a broader sense, the increasing possibilities within computational NLP, it becomes a pertinent question of how such methods can be used to provide a structured, quantitative way of narrative analysis of media texts.

This project demonstrates an NLP pipeline which ingests a dataset of news stories from prominent Swiss media outlets, extracts the narratives contained therein, clusters these narratives across different levels of granularity, and thereby provides an overview of which narratives are present within mainstream Swiss political-economic media discourse.

Concretely, the work addresses three broad questions:

- RQ 1: Can the fuzzy notion of a “narrative” be operationalized into a rigid, machine-processable structure that an LLM can reliably extract from a news text?
- RQ 2: When such structured narratives are embedded and clustered, which dominant narratives emerge in Swiss political-economic media coverage?
- RQ 3: Do different publishers emphasize distinct narratives, and how does this vary in sentiment and over time?

2. Data

2.1 Data Sources

The data used for this project was collected through the Swissdox database, which collects news texts from over 250 Swiss media outlets and makes them available to institutions and researchers.¹ Through the University of Zurich's Linguistic Research Infrastructure, this data can be accessed, queried and downloaded via API, provided one has an institutional login (which is true for this project through the University of Bern).²

The raw dataset created for this project consists of news articles published between 1st of June 2025 and 31st of May 2026, from 12 German-language Swiss media outlets, which belong to seven publisher groups, listed below.

¹ <https://swissdox.ch/>

² <https://swissdox.linguistik.uzh.ch/>

Table 1: Selected Publications

Publication Name	Swissdox Publication Code	Publisher
<i>20 Minuten Online</i>	ZWAO	TX Group
<i>Aargauer Zeitung</i>	AZM	CH Media
<i>Basler Zeitung</i>	BAZ	TX Group
<i>Berner Zeitung</i>	BZ	TX Group
<i>Blick Online</i>	BLIO	Ringier
<i>Die Wochenzeitung</i>	WOZ	Genossenschaft infolink
<i>Die Weltwoche</i>	WEW	Weltwoche Verlags AG
<i>Neue Zürcher Zeitung</i>	NZZ	NZZ-Mediengruppe
<i>Luzerner Zeitung</i>	LUZ	CH Media
<i>srf.ch</i>	SRF	Schweizerische Radio- und Fernsehgesellschaft SRG
<i>St. Galler Tagblatt</i>	SGT	CH Media
<i>Tages-Anzeiger</i>	TA	TX Group

2.2 Data Collection & Filtering

To collect the data as described above, the following query was sent to the Swissdox API, with the resulting TSV file then functioning as the raw data source for all subsequent steps.

```

query:
  sources:
    - BAZ
    - BZ
    - WEW
    - NZZ
    - LUZ
    - SRF
    - SGT
    - TA
    - WOZ
    - BLIO
    - ZWAO
    - AZM
  dates:
    - from: 2025-06-01
      to: 2026-05-31
  languages:
    - de
  result:
    format: TSV
    maxResults: 10000000
    columns:
      - id
      - pubtime
      - medium_code
      - medium_name
      - rubric
      - regional
      - doctype
      - doctype_description
      - language
      - char_count
      - dateline
      - head
      - subhead
      - article_link
      - content_id
      - content
version: 1.2

```

This query yielded an initial raw dataset of 245'080 articles.

This data was then filtered to limit the articles for analysis to specifically political or economic news topics to reduce noise and focus the data on the task at hand. Since the metadata

available per article varied across different publications in the data, this was done by using either the pre-given rubric of the article (e.g. “Politik” or “International”) where available or else by using keywords within the article’s URL. Additionally, the dataset was filtered to remove article headlines which occurred more than once (deduplication).

These steps resulted in a filtered dataset of 66’404 articles. Figure 1 shows the full reduction process; after narrative extraction (see Section 4.3), 44’786 articles where an extractable narrative was present remained. These articles form the basis for the normalization, clustering and labelling steps (see Sections 4.4 onwards) and for the subsequent analysis in the Streamlit application.

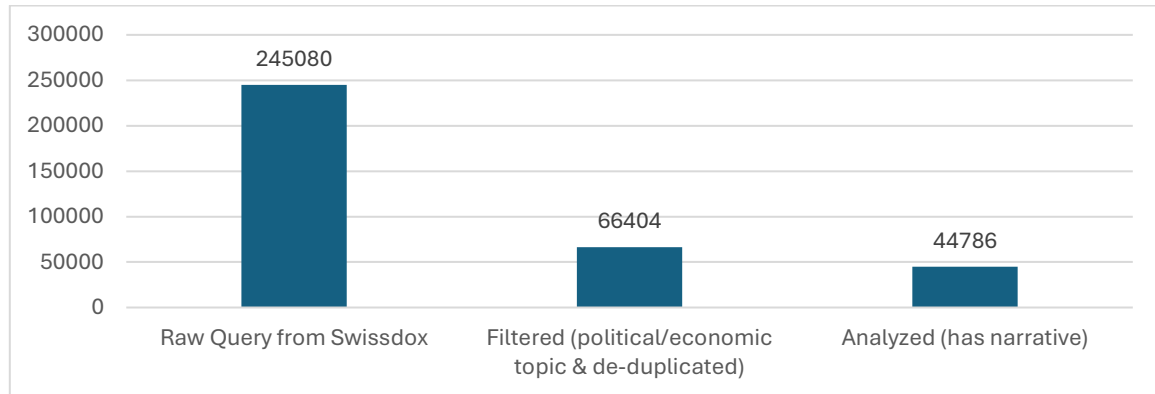


Figure 1: Article reduction process from raw query to analyzed set

2.3 Data Preprocessing & Feature Engineering

In further pre-processing steps, the filtered dataset was modified to drop unnecessary columns, made sure the content column contained all text of the article (by merging subhead elements to the content), and removed HTML tags from content.

Beyond the extracted narrative structure (Section 4), each article was enriched with features to be used in the exploratory analysis and the interactive application:

- Publisher: the parent company behind each outlet (see Table 1), as this will be the unit of analysis for comparison
- Sentiment: a score ranging from -1 to +1, computed using oliverguhr/german-sentiment-bert
- Emotion: four scores ranging from 0 to 1 (joy, anger, fear, sadness), computed using MilaNLProc/xlm-emo-t
- Linguistic complexity: a Flesch-Reading-Ease score

The complexity feature was computed and retained in the data tables but ultimately not surfaced in the final application, as it did not add interpretive value to the narrative comparison. All per-article metrics are stored alongside the cluster assignments in a single tidy article table that drives both the analysis below and the application.

3. Exploratory Data Analysis

The filtered corpus comprises 66,404 articles from 7 publishers. Volume is highly uneven across publishers: the three highest-volume online publishers — Ringier (28.7%), TX Group (25.5%) and SRG (22.6%) — together account for roughly three quarters of the corpus, while weekly titles such as Weltwoche (2.7%) and infolink's Die Wochenzeitung (1.0%) contribute far fewer articles.

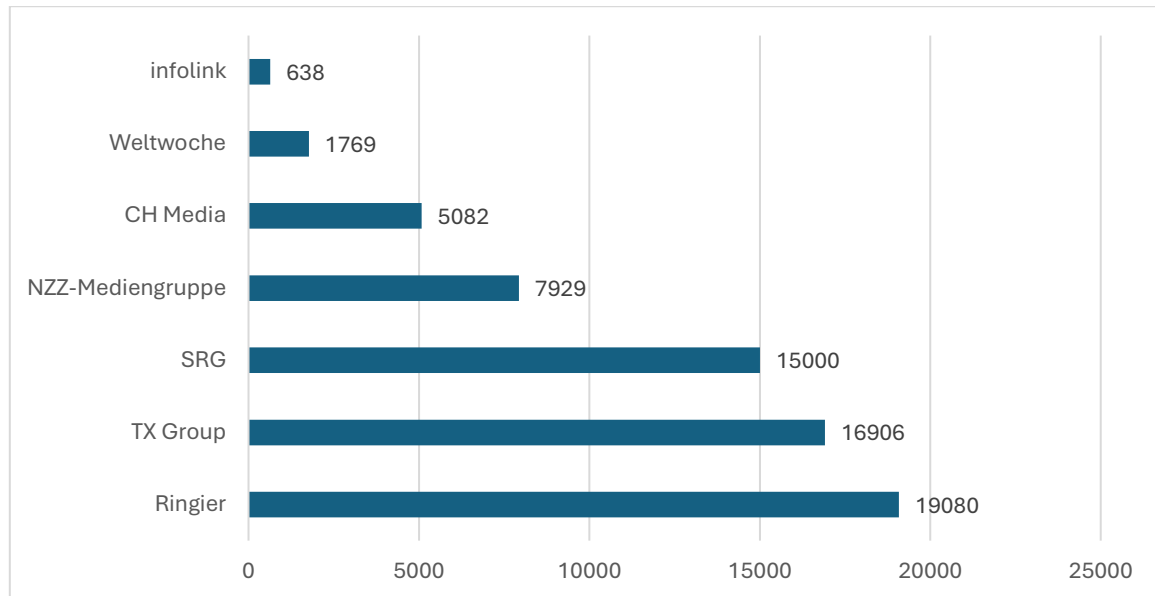


Figure 2: Filtered article count per publisher

This pattern of articles volumes is more or less consistent over time, as the graph below shows. There is a noticeable drop in published articles for all publishers over the festive period at the end of December, which is reasonable to expect.

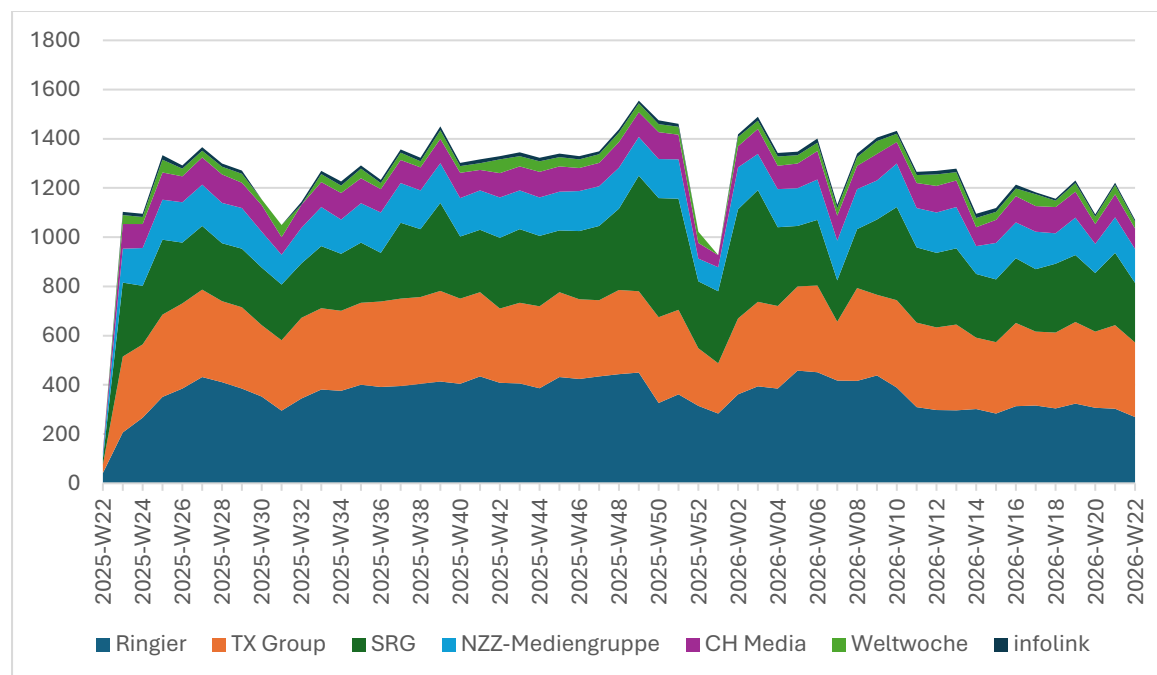


Figure 3: Filtered article count per publisher, over time

As for the additionally engineered features, as the figures below show, average overall sentiment was negative across all publishers (to be expected for news articles), but with quite some variation: Weltwoche clearly shows the most negative coverage, while SRG is the most neutral.

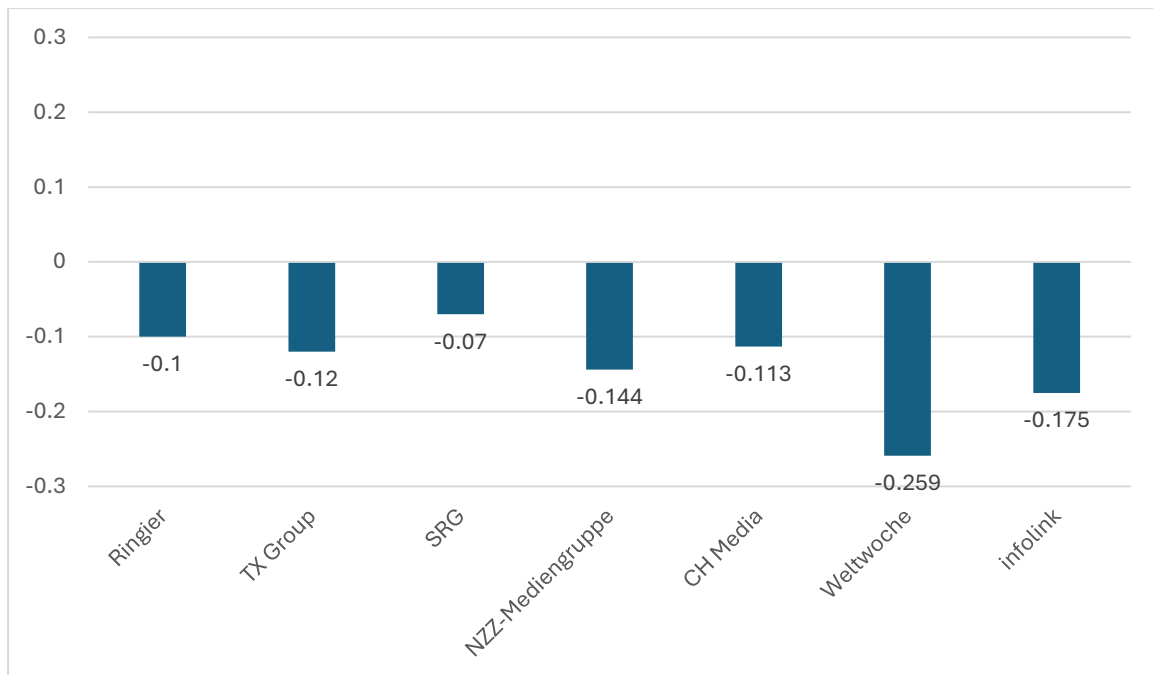


Figure 4: Average sentiment per publisher

There is also some variation in the averages of the four emotion scores, but there are fewer clear outliers there. Across all publishers, anger is the dominant emotion, while joy and sadness are roughly equivalent, and fear is least pronounced.

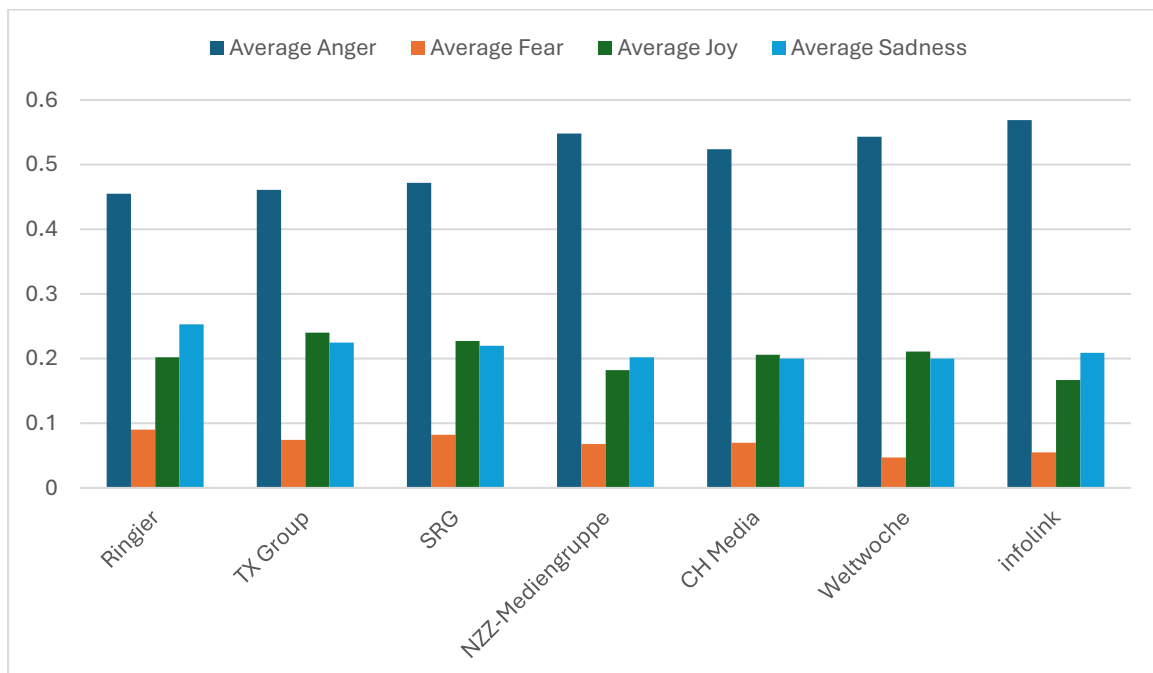


Figure 5: Average emotion scores per publisher

4. NLP Analysis

4.1 Defining Narrative Structure

The first core challenge of this project, corresponding to RQ 1, was to determine a way with which the somewhat fuzzy concept of a “narrative” could be operationalized into a rigid structure fit for natural language processing tasks.

The main theoretical framework eventually used to derive such a structure was Greimas’ (1966) actantial model. In this framework, a narrative is represented as revolving around some sort of “quest”, which is a common element of both fictional as well as real-world storytelling. The framework distinguishes six roles (“actants”) which are jointly representing the core content of the narrative: subjects, objects, helpers, opponents, senders, receivers.

Table 2 below shows these different actant roles, their descriptions as well as two ideal-type examples for each – one fictional, one real.

Table 2: Actant Roles and Examples

Actant	Description	Fictional Example	News Example
<i>Subject</i>	<i>protagonist actively pursuing the object</i>	- Frodo	- Government of Ukraine
<i>Object</i>	<i>what is desired, sought, or at stake</i>	- Destruction of the ring	- Expulsion of Russian forces from Ukrainian territory
<i>Sender</i>	<i>mandate or value authorizing the quest</i>	- Council of Rivendell	- National sovereignty - Peace & Security
<i>Receiver</i>	<i>ultimately benefits or suffers from the quest outcome</i>	- Free peoples of Middle Earth	- Population of Ukraine - Security of Europe
<i>Helper</i>	<i>enables the quest</i>	- The Fellowship - Gollum	- NATO
<i>Opponent</i>	<i>obstructs the quest</i>	- Sauron - Saruman - Gollum	- Russia

Actant roles can be filled by multiple entities at the same time, or none at all. Furthermore, an entity can occupy multiple roles simultaneously (e.g. Gollum in the Lord of the Rings can be seen as both a helper and an opponent).

Importantly, such a “quest” framing is not equivalent to a simple statement-of-fact or a descriptive representation of information – rather, the same situation may be represented by many different narratives implying different roles for involved entities. Furthermore, not every text necessarily contains a narrative in this sense (a weather forecast, for example, would most likely not contain a quest-styled narrative).

4.2 Defining Narrative Slot Fillers

Given this overall structure, each narrative offers a maximum of six roles (“slots”) to be filled by different entities (“slot fillers”). To allow for meaningful analysis, a next relevant question then becomes to define a framework within which the different potential slot fillers can be categorized.

Drawing from different schools of thought (e.g. the state/market/civil-society distinction by Cohen & Arato, 1992; cleavage theory by Lipset & Rokkan, 1967) as well as some personal imagination and eventual refinement with LLM assistance, I iteratively developed and finally settled on the following hierarchical categorization of potential slot fillers:

At the highest level, there is a **class** distinction to be made on whether the slot filler is a distinct actor (an entity with its own agency), an abstract force (a constraining or enabling factor or object with no agency of its own), or an ideological frame (a normative worldview). Hypothetically, any of the three classes can fill any of the slots, but in real life, some configurations are much more likely (e.g. subjects are typically actors, and senders are often ideological frames).

In a second level, for each class, there are defined **categories**. For actors, the categories represent different sectors of actors, with the types being specifically chosen to work from a Swiss political perspective (e.g. differentiation between domestic state actors and foreign actors). For abstract forces, there are two categories which indicate whether the slot filler is a general structural condition or an instrument that can be deployed by an actor. For ideology, the categories are derived from different axes of political ideologies (e.g. market liberalism vs. redistribution), as well as a category of legitimation modes being added.

Finally, on a third level, I define the **specific types** of slot fillers. These are more specific sub-categories to the second-level categories. In general, they are simply more specific versions of the second-level categories. Table 3 below gives an overview of all slot filler options.³

Table 3: Slot Filler Categorization Framework

Classes	Categories	Specific Types
<i>actor</i>	<i>domestic_state_actor</i>	<i>executive · legislature · judiciary · regulatory_agency · security_enforcement · public_administration</i>
	<i>market_economic</i>	<i>corporation_firm · financial_institution · industry_trade_association · labour_organisation · individual_entrepreneur</i>
	<i>civil_society</i>	<i>ngo_advocacy · social_movement_initiative · political_party · religious_institution · civic_association · legal_professional</i>
	<i>media_knowledge</i>	<i>journalist_media_outlet · think_tank · academic_scientific · individual_expert · polling_statistics</i>
	<i>general_public</i>	<i>citizens_voters · consumers_households · workers_employees · affected_community · individual_unaffiliated</i>
	<i>supranational_actor</i>	<i>eu_institution · intergovernmental_organisation · intl_humanitarian · intl_security_alliance · intl_financial_institution</i>
	<i>foreign_actor</i>	<i>foreign_executive · foreign_legislature · foreign_judiciary_regulatory · foreign_public_administration · foreign_political_party · foreign_civil_society · foreign_military · foreign_paramilitary</i>

³ All specific types and their concrete definitions can be inspected within the LLM extraction prompt file used in the extraction Jupyter notebook.

<i>abstract_force</i>	<i>structural_condition</i>	<i>institutional_framework</i> · <i>economic</i> · <i>demographic</i> · <i>technological</i> · <i>geopolitical</i> · <i>environmental</i> · <i>cultural_historical</i>
	<i>deployed_instrument</i>	<i>legal_instrument</i> · <i>policy_instrument</i> · <i>economic_instrument</i> · <i>communicative_instrument</i> · <i>technological_instrument</i>
<i>ideology</i>	<i>economic_axis</i>	<i>market_liberal</i> · <i>redistributive</i>
	<i>cultural_axis</i>	<i>cosmopolitan</i> · <i>communitarian</i>
	<i>authority_axis</i>	<i>libertarian</i> · <i>authoritarian</i>
	<i>legitimation_mode</i>	<i>technocratic</i> · <i>religious</i> · <i>ecological</i>

Given this categorization framework, I expected to capture differences in narratives in a structured and generic way.

For example, an imaginary narrative might in this way be described as a *domestic_legislative* actor (subject) pursuing a *policy_instrument* (object), motivated by *redistributive* ideology (sender), against the opposition of a specific *political_party* actor (opponents).

A narrative characterized in this way is a priori independent of time, place or positive/negative framing. It was my hope for further analysis that in this way, I would be able to find common narrative structures across many different contexts.

Additionally, I also situated each slot filler within an “entity domain”, which encapsulates within which sphere this entity is situated (e.g. political power, social welfare, etc.). However, I ended up not using this information in the visualizations in the final app.

4.3 Narrative Slot Extraction

To test RQ 1, based on this framework, an extraction system prompt was designed to task an LLM with providing, for each article in the dataset, a structured JSON object containing the relevant information on the main narrative within the article.

The LLM is tasked with five subsequent steps:

1. Determine whether there is a narrative present or not
2. Generate a one-sentence-summary of the narrative
3. Fill all narrative slots with entities (if present)
4. Classify each entity according to the framework of Table 3
5. Emit the result as JSON.

The model used was Anthropic’s Claude Haiku 4.5 (claude-haiku-4-5-20251001) via the Batches API.

Extraction quality was assessed manually by spot-checks and repeated across multiple sample runs, in tandem with the refinement of the slot filler taxonomy as well as the prompt, before committing on the final full run across the full preprocessed dataset.

A narrative was found by the model in 67.4% of filtered articles, yielding 44’786 structured narratives. The remaining 21’618 articles were judged to contain no quest-structured narrative (e.g. pure news flashes, quizzes etc.) and are excluded from the rest of the pipeline.

4.4 Normalization of Slot Filler Labels

Free-text entity labels produced by the LLM contain many surface variants of the same underlying entity (spelling, abbreviation, language). To canonicalise them, labels were embedded with a multilingual sentence encoder (intfloat/multilingual-e5-large) and near-duplicate labels were merged, producing a normalised entity vocabulary that is used both for clustering and for the entity-level search in the application.

Due to the sheer number of entity labels produced (over 200'000), it was found retroactively that more precise instructions in the initial extraction prompt would probably have been helpful in limiting the different surface forms produced by the LLM during narrative extraction. However, due to the token cost associated with running a full extraction on 67'000 articles, it was decided to not re-run with an adapted prompt.

4.5 Macro-, Meso- and Micro-Clustering

The next step was to cluster the 44'786 narratives in a meaningful way in order to assess the underlying structures. Since a one-size-fits-all approach would entail a very impactful choice of granularity, it was decided to cluster hierarchically, using UMAP and HDBSCAN, across three levels of granularity.

Features used in clustering fell into three categories: structural – a binary matrix of the slot filler categories and specific types (see Table 3), semantic – an embedding of the one-sentence narrative description (using intfloat/multilingual-e5-large), and slot – mean-pooled embeddings (same embedding model) of the entity descriptions that filled the narrative slots. For each level, different clustering weights were given to the different features within the narrative structure.

In the first, macro level, clustering was driven almost entirely by the structural block, with the semantic block down-weighted to 0.1 and the slot block omitted. The aim at this level was to recover broad narrative archetypes defined by who acts on whom in what capacity, independent of the concrete topic. Clustering was performed over the entire corpus with a large neighbourhood ($n_neighbors = 200$) and a high minimum cluster size (150), yielding 17 macro-narratives.

In the second, meso level, each macro-narrative was subdivided separately, this time giving the semantic block the dominant weight (0.7). Here the goal was to split each structural archetype into topically coherent sub-narratives – content variants that share the same actantial skeleton but differ in subject matter. Because macro-narratives vary greatly in size, the neighbourhood and minimum-cluster-size were set proportionally to each parent's article count (bounded within fixed minima and maxima) rather than as fixed constants.

In the third, micro level, each meso-narrative was in turn subdivided, now introducing the slot block (weight 0.1) alongside only a minimal semantic contribution (0.05). At this finest level the intention was to separate distinct perspectives on the same topic – narratives that differ in the specific actors and forces occupying the actantial roles (e.g. which party is cast as helper or opponent). Again, the parameters scaled with the size of each parent cluster.

Parameters were determined iteratively in order to maximize cluster coherence and reduce noise. A grid search was run at each level over the UMAP neighbourhood, the HDBSCAN minimum cluster size and minimum samples (and, at the meso and micro levels, over the feature weights), scoring each configuration on the resulting number of clusters, the share of articles left as noise, and a coherence measure based on the concentration of a cluster's most frequent entities. The chosen configurations were then validated with an automated check that flags "catch-all" clusters and clusters falling below a minimum viable size.

Because HDBSCAN does not force every narrative into a cluster, a share of narratives was left unassigned as noise at each level, and this share grew as the granularity increased. At the macro level 91.7 % of the 44'786 narratives were assigned to one of the 17 macro-

narratives, leaving 8.3 % (3'711 narratives) as noise. Clustering became more selective at the finer levels: of the macro-assigned narratives, 77.2 % were further placed into one of 226 meso-narratives, and of those, 75.5 % were placed into one of 351 micro-narratives. Cumulatively, this meant that 70.8 % of all narratives carried a meso label and 53.5 % carried a micro label; the remainder were retained at the coarser level to which they had been assigned but were not subdivided further.

4.6 Cluster Labelling

In order to allow for a more intuitive understanding of cluster content, each cluster at every level was assigned a concise human-readable label. Rather than labelling manually, a large language model (Anthropic Claude Sonnet, temperature 0.1) was used to summarize each cluster from a structured evidence bundle. For every cluster this bundle comprised a set of representative articles – sampled in proportion to cluster size and drawn from those closest to the cluster centroid – together with the cluster's most frequent slot fillers and its dominant entities. From this evidence the model was prompted to return a short title and a description of at most twelve words.

To keep the labels consistent across the hierarchy, the prompts were made context-aware: a meso-narrative was labelled with its parent macro title supplied as context, and a micro-narrative with both its macro and meso ancestors, so that finer labels read as refinements of their parents. In addition, recently assigned sibling labels were passed to the model to discourage near-duplicate names within the same branch. The three levels were labelled independently and then merged into a single hierarchical label table linking every macro-, meso- and micro-narrative to its title and description.

Labeling prompts were also developed iteratively, with labeling runs on smaller subsets of the data, before committing to the full run once label quality was assessed to be sufficient. The issue with the diverse range of entity labels (see 4.4) surfaced again here, since it can be presumed that gauging the main elements of a narrative in order to label it would have been more straightforward if entity labels for slot fillers would have been more standardized. Labels for certain narratives are therefore to be taken with the proverbial grain of salt.

4.7 Visualization

Rather than just presenting fixed graphs, for explorative visualization of the data, an interactive Streamlit app was developed (relying heavily on Claude for the UI code development, but with a decently large number of “human-brain hours” spent on optimizing workflow and functionality specifications).

The app is accessible directly from <https://swiss-narrative-analysis.streamlit.app/>.

It features two main modes of analysis: On the one hand, *narrative* mode enables the selection, detailed inspection and comparison of clustered narratives. This narrative view also enables the formation of ad-hoc narrative clusters based on user input either via semantic search of narrative propositions (which are compared to the one-sentence narrative descriptions by the LLM generated at extraction time), or via keyword search of entities present in the narrative slots. All narrative “cards” can be filtered according to publisher or week of publication and the filtered sets compared with one another.

On the other hand, *entity* mode enables browsing different entities via free-text search in the existing slot fillers. In this view, one can compare what narrative roles (slots) and emotional profile a given entity has within the dataset and across different publishers.

More detailed explanations and guidance on how to use the app are contained directly in the app itself.

5. Results & Discussion

Given that the main presentation of the NLP pipeline results is done via the Streamlit app, this section will be limited to providing a general illustrative overview of the 17 macro clusters and how they are represented across publishers. For more fine-grained analysis, it is advised to use the app.

As an illustration, the figure below shows a 2D UMAP projection of all articles, colored by macro-cluster. It should be noted that the dispersed nature of clusters in 2D is to be expected, since the actual clustering was done on a 50-dimensional UMAP reduction, not in 2D.

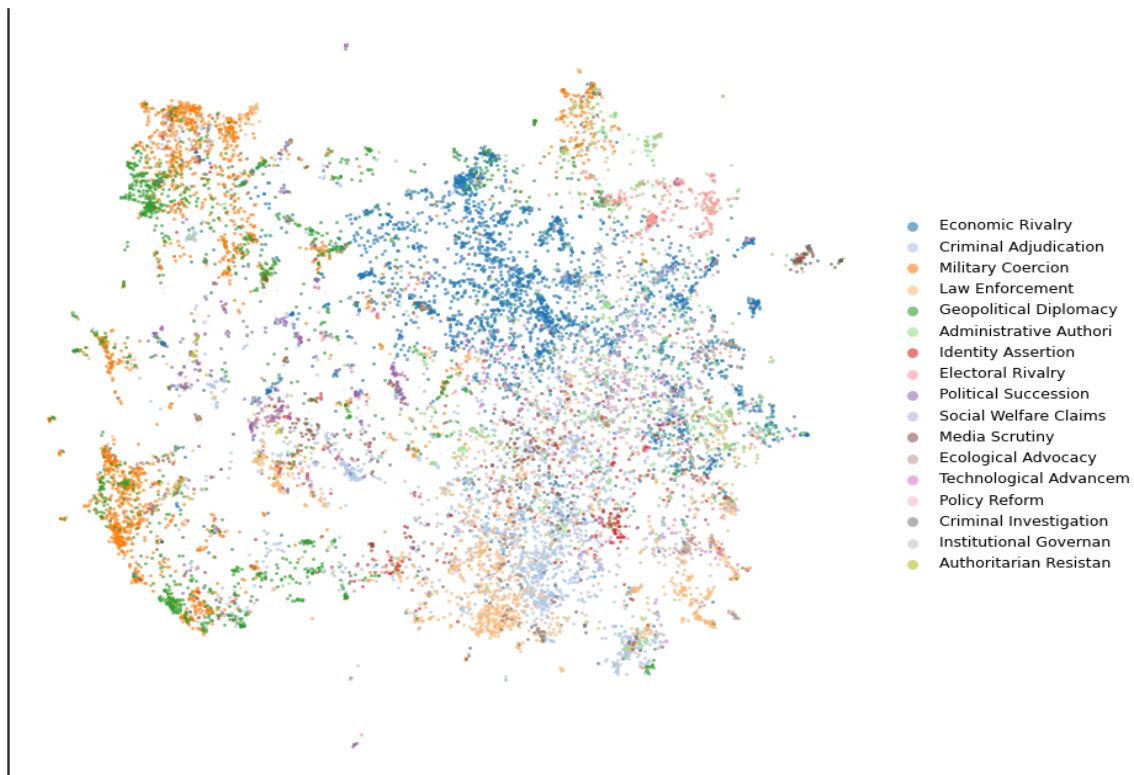


Figure 6: 2D UMAP projection, colored by macro-cluster

Table 4: Article count per macro-cluster

Macro Cluster	Economic Rivalry	Criminal Adjudication	Military Coercion	Law Enforcement	Geopolitical Diplomacy	Administrative Authority
Count	9560	6077	5830	4580	4076	3309
Macro Cluster	Identity Assertion	Electoral Rivalry	Political Succession	Social Welfare Claims	Media Scrutiny	Ecological Advocacy
Count	1151	1142	1128	1119	1049	758
Macro Cluster	Technological Advancement	Policy Reform	Criminal Investigation	Institutional Governance	Authoritarian Resistance	
Count	405	282	222	208	190	

Overall, there are some patterns emerging: The biggest single cluster deals with economic rivalry, where a lot of articles are presenting narratives about Trump's tariffs and reactions to them, for example. Furthermore, there are some clusters about military conflicts and geopolitics, which is to be expected given the ongoing wars in Ukraine and the Middle East. There are further clusters on criminal cases and law enforcement, which also seems sensible for the rubrics that were chosen for the initial dataset. Some further clusters on diverse topics also exist, which all seem reasonable for the Swiss political-economic media landscape.

When looking at the relative prevalence of clusters across different publishers, there are some interesting observations on the difference between publishers. NZZ and CH Media, for example, focus most on economic rivalry. Ringier (i.e. Blick) exhibits more tendency to cover crime and law enforcement stories than the average. Weltwoche shows the highest value on media scrutiny, which may be interpreted as it positioning itself more as an outsider to the media mainstream.

Table 5: Relative frequencies of clusters across publishers

Macro narrative	Overall	Ringier	TX Group	SRG	NZZ	CH Media	Weltwoche	infolink
Economic Rivalry	21.3	24.2	19.4	18.5	32.2	31.0	16.6	14.8
Criminal Adjudication	13.6	16.8	17.7	12.2	11.0	12.9	13.3	26.3
Military Coercion	13.0	12.4	12.6	18.3	15.8	12.5	8.9	8.9
Law Enforcement	10.2	16.3	12.0	10.2	5.5	7.8	3.1	5.3
Geopolitical Diplomacy	9.1	7.2	8.6	13.9	10.8	8.8	14.7	7.9
Administrative Authority	7.4	7.9	7.5	8.4	7.8	9.6	8.7	5.7
Identity Assertion	2.6	1.9	4.9	2.0	2.3	1.8	7.1	4.0
Electoral Rivalry	2.5	1.8	2.8	2.5	3.7	3.6	8.4	4.7
Political Succession	2.5	1.6	3.0	3.1	3.8	2.8	3.1	3.0
Social Welfare Claims	2.5	3.0	3.0	3.1	1.5	2.4	0.7	5.7
Media Scrutiny	2.3	2.2	3.5	1.5	1.8	2.9	10.5	4.3
Ecological Advocacy	1.7	1.8	1.5	2.8	0.8	1.4	2.9	6.3
Technological Advancement	0.9	0.8	1.3	1.1	0.8	0.6	1.1	0.2
Policy Reform	0.6	0.7	0.5	0.9	0.5	1.0	0.0	0.6
Criminal Investigation	0.5	1.0	0.7	0.3	0.1	0.3	0.0	0.0
Institutional Governance	0.5	0.3	0.4	0.6	1.0	0.4	0.2	0.4
Authoritarian Resistance	0.4	0.3	0.6	0.5	0.5	0.4	0.7	2.0

Further analysis could be done on the prevalence of different emotions and overall sentiment across publishers, but these insights can also be made by exploring the Streamlit app, which is why they are omitted from this report.

6. Conclusion

This project set out to operationalize narrative analysis of news text and to apply it at scale to Swiss political-economic coverage. Using Greimas's (1966) actantial model and a three-level slot-filler taxonomy, an LLM extracted structured narratives from 67.4% of 66,404 filtered articles; embedding and clustering these structures produced 17 macro-, 226 meso- and 351 micro-narratives (RQ1, RQ2). The landscape is dominated by economic rivalry, criminal adjudication and military coercion, is mildly negative throughout with anger the most prominent emotion, and shows measurable, interpretable differences in emphasis across publishers and over time (RQ3). The accompanying application turns these results into an explorable map of Swiss news narratives.

However, the results need to be read with several sources of uncertainty in mind, which mirror the choices made at different stages of the established NLP pipeline.

First, the input data and its filtering: The corpus is restricted to twelve German-language outlets and one year, so findings characterize this slice of the Swiss media landscape rather than all Swiss media in general. Furthermore, the political/economic topic filter relies partly on URL keywords where rubrics are missing, which introduces some classification noise.

Second, the narrative structure and slot filler taxonomy: the narrative structure to be operationalized and its slot-filler taxonomy were designed by a single author and reflect specific theoretical choices; a different taxonomy could yield very different typings. For example, it was found that the specific type of “policy instrument” was widely present in slot fillers and may therefore benefit from further splitting.

Third, extraction reliability: narrative structure is assigned by an LLM, and the same article could plausibly be told through more than one narrative. Using a different model, or slightly altering the prompt, may have noticeable implications for results. A more systematic inter-model or human-agreement study, which was beyond this project, would be needed to quantify this deviation.

Fourth, clustering stability: Generally, while it was found that small changes to the clustering parameters were not producing wildly different structures, the overall logic behind clustering three different levels with the specified different weights for feature types may be questioned and potentially adapted for the better.

Beyond these obvious candidates for further improvement, the project still managed to achieve its original goal: From a large dataset of news articles, a structured NLP pipeline and a related app provide an interactive way to browse narratives in Swiss media. The general functioning and logic behind the pipeline is theoretically domain-agnostic, and with different data, adapted prompts, or different models, may be suitable for use in a variety of contexts.

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